

Faculty of SCET
Swinburne University of Technology
SWE30003 Software Architectures and Design
Tutorial Week 3
(User Tasks and Requirements)

Hotel Information System

1. Considering the notions of Product and Domain around Lecture 2, Slide 25), is the receptionist part of the product or part of the domain? Please explain!

=> From my opinion, the receptionist is of the domain because the receptionist is an external human actor whose real-world tasks define the "Domain-level requirements" that the system must support.

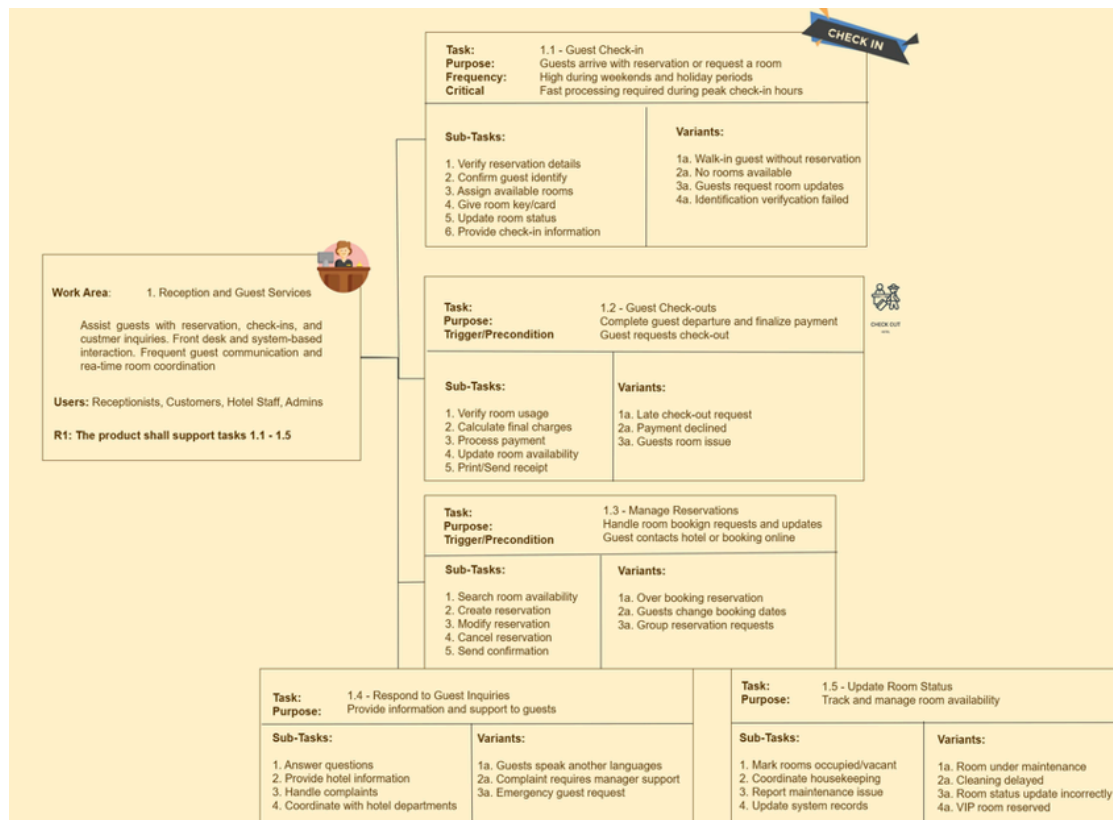
2. Given the context of the hotel information system as discussed in Lecture 2, describe the *Check-Out* task in the Task & Support style.

Task: Purpose: Trigger/Precondition	1.2 - Guest Check-outs Complete guest departure and finalize payment Guest requests check-out	 CHECK OUT HOTEL
Sub-Tasks: 1. Verify room usage 2. Calculate final charges 3. Process payment 4. Update room availability 5. Print/Send receipt	Variants: 1a. Late check-out request 2a. Payment declined 3a. Guests room issue	

3. Identify the typical tasks that a Hotel Information System must support, create a corresponding *workflow*, and visualize the resulting workflow using a suitable notation, and describe it as a high-level task. => Hoàng

- Typical tasks supported by a hotel information system:

1. **Room reservation:** Checking availability, recording guest details, and securing a booking.
2. **Check-In:** Verifying reservation, assigning a physical room and issuing a key.
3. **Service Log & Stay Management:** Recording additional charges to the guest portfolio.
4. **Check-Out:** Consolidating expenses, calculate & processing the final payment and update room status to "Dirty/Vacant".
5. **Room Cleaning & Maintenance:** After cleaning up staff update the room status "Clean/Available".



High-Level Task: Manage Guest Stay Lifecycle

- **Goal:** Successfully transition a guest from initial arrival/booking through their entire stay to final payment and room vacation, while ensuring hotel operational resources (rooms, billing records) remain consistent.

4. What are tacit requirements? Is it possible to perform a validation check on tacit requirements? => Dăng

=> Tacit requirements (or tacit demands) represent requirements that are unstated by stakeholders because they are taken for granted, implied, or deeply embedded in daily habits.

Yes, it is possible to perform a validation check on tacit requirements

- Scenarios and Stories: Running a "Sanity Check" by tracing real-world user paths to see if your documented tasks actually cover every operational step.
- CRUD Completeness Check: Mapping your tasks against domain entities (Guest, Room, Booking). If an entity can be created but has no task to read, update, or delete it, a tacit requirement has been missed.
- Screen Prototypes / Sample Solutions: Showing mockups to stakeholders so they can physically point out missing fields or unstated workflows.

5. The domain model for the hotel information system (around Lecture 2, Slide 31) is incomplete. For example, in most hotels, rooms are by room *type* (single,

double, family etc.). Furthermore, overbooking of rooms is not supported, either. Extend the domain model so that these extensions can be covered. => Hiểu

6. The *Check-In* task (around Lecture 2, Slide 35) handles two situations: => Đăng

a) The guest has booked in advance;

b) The guest has not booked in advance, but hopes there is a free room available.

Do the following:

1) Split the task description into two task descriptions *Check-In a Booked Guest* and *Check-In a Non-Booked Guest*.

=>

Check-In:

- Trigger/Precondition: A guest arrives with an existing reservation record in the system.
- Sub-tasks:
 1. Ask for guest identification or booking reference number.
 2. Retrieve the existing booking record from the system.
 3. Verify guest details against the record.
 4. Allocate a physical room matching the booked type.
 5. Update room state to "occupied" and deliver the key card.

Check-out:

- Trigger/Precondition: A guest arrives without a reservation; the hotel must have vacant rooms.
- Sub-tasks:
 1. Check system room states to find available free rooms.
 2. Present room types and pricing/bargaining options to the guest.
 3. Manually record all new guest profile information (Name, Passport, Address).
 4. Create a new active stay record and start the account.
 5. Allocate the chosen room, update status to occupied, and deliver the key.

2) Discuss the difference between the one-task solution and the two-task one. Which approach is more suitable?

=> Compare:

- One-Task Solution: Bundles both paths into a single description using complex conditional rules (if/else branching). It leads to cluttered UI layouts trying to handle two different jobs at once.
- Two-Task Solution: Splits them into two clean, linear workflows. It results in a cleaner UI dashboard layout (e.g., a "Find Booking" button vs. a "New Walk-in" button).

According to the lecture guidelines, the one-task approach (postponing the split) is preferred during early domain analysis. Keeping them together ensures you comprehensively capture all related business rules and high-level workflows of the work area before breaking them into separate software application use cases.

7. Many hotels apply a policy of *overbooking*, i.e. they book more rooms than the hotel actually has. In practice, not all booked guests turn up and there is no real problem. However, in case too many guests turn up, an emergency solution needs to be found (e.g., allocating guest to a different hotel). Please extend the *Check-In* task to handle this situation. => Hoàng

- Check-In task With OverBooking Exception

Task 1.1: Guest Check-In

Purpose: Process arriving guests with or without prior reservations, allocate a physical room and handle operational capacity with care.

Trigger/Precondition: Guest arrives at the reception desk requesting to check in.

Frequency: High (especially during afternoon windows, weekends, and holiday periods in a year).

Critical: Fast processing required during peak hours to minimize queue times. Real-time accuracy is mandatory to prevent duplicate room assignments.

Sub-Tasks:

Verify Reservation: Request and verify guest identification, booking reference, or check for walk-in availability.

Assess Capacity: Query the system for physical room availability matching the reserved room type.

Handle Exceptions (Overbooking): If physical inventory for the requested room type is at zero (0), evaluate the compensation and reallocation protocols to other types of room or to partner/sister hotel .

Assign Room: Allocate a specific vacant physical room number to the guest profile.

Issue Access: Program and hand over physical room key cards.

Update State: Update the room status in the central database to "Occupied".

Complete Process: Provide check-in info and print/email the registration receipt.

8. Any questions from weekly submissions? => Currently is not